

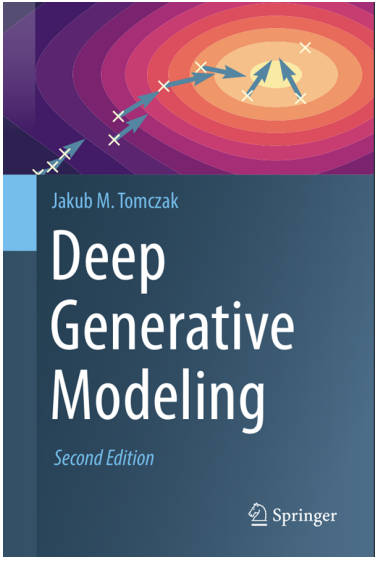
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Advanced Machine Learning - COMP8221

Deep Generative Models - 1

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Recommended textbook



Outline

Probabilities and Distributions

Probabilistic Graphical Models

Continuous Optimization



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Probabilities and Distributions

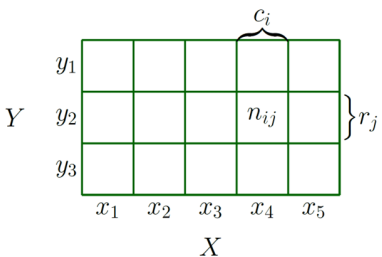
Random variables

- Random variables $\mathbf{x}$: its outcome/value might be changed/uncertain, **randomness**
- Sample space Ω: random variable $\mathbf{x}$'s potential values
- E.g., flip a coin ($\Omega = \{\text{Positive, Negative}\}$), tomorrow's temperature $\Omega = \mathbb{R}$, stock market price $\Omega = \mathbb{R}$, pixels in an image if taken a picture $\Omega = \{0, 1, 2, 3, \ldots, 255\}$
- Observations (e.g.) $\mathcal{D} = \{\mathbf{x}_1 = 234, \mathbf{x}_2 = 12, \ldots\}$: the detail values of random variables in a sequence of observations
- Question: What are the chances of all the potential values? What if some values are more likely than other values?

Probability distribution

- E.g. Pixels values, some pixels values are more likely than other pixel values
- We use a probability distribution $p(\mathbf{x})$ to represent the likelihood value
- $\mathbf{x} \sim p(\mathbf{x})$ denotes we sample an $\mathbf{x}$ from the probability distribution $p(\mathbf{x})$.
- Requirements of $p(\mathbf{x})$:
  - If $\mathbf{x}$ is discrete-valued, $0 \leq p(\mathbf{x}) \leq 1$; if $\mathbf{x}$ is continuous-valued, $p(\mathbf{x}) \geq 0$.
  - The probabilities are sum to 1. $\sum_{\mathbf{x} \in \Omega} p(\mathbf{x}) = 1$ or $\int_{\mathbf{x} \in \Omega} p(\mathbf{x}) d\mathbf{x} = 1$,
- Question: pixel values, $p(\mathbf{x} = 0) + p(\mathbf{x} = 1) + \ldots + p(\mathbf{x} = 255) = ?$

- The joint discrete probability is define as (n_{ij} is the number of events with $\mathbf{x} = x_i, \mathbf{y} = y_j$)

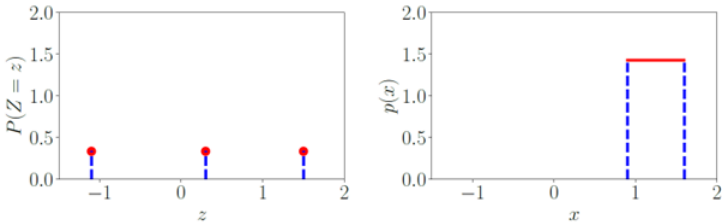


$$P(\mathbf{x} = x_4, \mathbf{y} = y_2) = \frac{n_{42}}{\sum_{i,j} n_{ij}} \qquad (1)$$

Continuous probabilities

- Probability density function (pdf): A function $f(\mathbf{x})$ is a pdf if: (1), $f(\mathbf{x}) \geq 0$; (2) $\int f(\mathbf{x}) d\mathbf{x} = 1$.
- $P(a \leq X \leq b) = \int_a^b f(x) dx$
- Cumulative distribution function (cdf): $F_X(x) = P(X \leq x)$

Contrasting discrete and continuous probabilities



(a) Discrete distribution

(b) Continuous distribution

Type	"Point probability"	"Interval probability"
Discrete	$P(X = x)$ Probability mass function	Not applicable
Continuous	$p(x)$ Probability density function	$P(X \leqslant x)$ Cumulative distribution function

Sum rule, product rule

- Sum rule: $p(\mathbf{x}) = \begin{cases} \sum_{\mathbf{y}\in\mathcal{Y}} p(\mathbf{x},\mathbf{y}), & \text{if } \mathbf{y} \text{ is discrete;} \\ \int_{\mathcal{Y}} p(\mathbf{x},\mathbf{y}) d\mathbf{y}, & \text{if } \mathbf{y} \text{ is continuous.} \end{cases}$
  From joint distribution to marginal distribution: $p(x_i) = \int p(x_1, \ldots, x_D) d\mathbf{x}_{\setminus i}$
- Product rule: $p(\mathbf{x},\mathbf{y}) = p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|\mathbf{y})p(\mathbf{y})$
  From joint distribution to conditional distribution.
- $p(\mathbf{y}|\mathbf{x})$: conditional probability, given $\mathbf{x}$, what is the probability of $\mathbf{y}$
- E.g., if $\mathbf{x} = \text{cat}, \mathbf{y} = \text{pixels}$, $p(\mathbf{y}|\mathbf{x})$ the probability of pixels for a cat, $p(\mathbf{y})$ the probability of pixels for all the animals, persons, buildings, etc.
- Independence: $p(\mathbf{x},\mathbf{y}) = p(\mathbf{x})p(\mathbf{y})$. I.e., $p(\mathbf{y}|\mathbf{x}) = p(\mathbf{y})$, $p(\mathbf{x}|\mathbf{y}) = p(\mathbf{x})$
- Question, $\mathbf{x}$: pixels; $\mathbf{y}$: weather. What is $p(\mathbf{x}|\mathbf{y})$? what is $p(\mathbf{y}|\mathbf{x})$?

- A common assumption for a dataset $\mathcal{D}$ is *identically* and *independently distributed* (*i.i.d.*):
  - all $\mathbf{x}_n \in \mathcal{D}$ follow the same distribution, i.e., $p(\mathbf{x}_n)$
  - all $\mathbf{x}_n$ are independent from others, i.e., $p(\mathbf{x}_1, \ldots, \mathbf{x}_N) = \prod_{n=1}^{N} p(\mathbf{x}_n)$.
- Question: Flipping a coin
  - identically: ?
  - independently: ?

- Assume we have prior knowledge $p(\mathbf{x})$ about unknown random variable $\mathbf{x}$ and relationship $p(\mathbf{y}|\mathbf{x})$ between $\mathbf{x}$ and observed random variable $\mathbf{y}$, the *Bayes' Theorem* is written as:

$$p(\mathbf{x}|\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{x})p(\mathbf{x})}{p(\mathbf{y})} \tag{2}$$

where $p(\mathbf{x}|\mathbf{y})$ is the posterior, $p(\mathbf{y}|\mathbf{x})$ is the likelihood, $p(\mathbf{x})$ is the prior, and $p(\mathbf{y})$ is the evidence.

- The *mean* of a random variable X with $\mathbf{x} \in \mathbb{R}^D$ is an average and is defined as:

$$\mathbb{E}_X[\mathbf{x}] = \begin{bmatrix} \mathbb{E}_{X_1}[x_1] \\ \vdots \\ \mathbb{E}_{X_D}[x_D] \end{bmatrix} \in \mathbb{R}^D \tag{3}$$

where

$$\mathbb{E}_{X_d}[x_d] := \begin{cases} \int_{\mathcal{X}} x_d p(x_d) dx_d, & \text{if } X \text{ is a continuous random variable;} \\ \sum_{x_i \in \mathcal{X}} x_i p(x_d = x_i), & \text{if } X \text{ is a discrete random variable;} \end{cases} \tag{4}$$

- Question: flipping a coin, head $= 1$, tail $= 0$, $p(\text{head} = 1) = 0.5$, what is the mean value? What if $p(\text{head} = 1) = 0.7$?

Covariance

- Univariate covariance of two univariate random variable $X, Y \in \mathbb{R}$ is calculated as:

$$\text{Cov}_{X,Y}[x, y] := \mathbb{E}_{X,Y}\left[(x - \mathbb{E}_X[x])(y - \mathbb{E}_Y[y])\right] = \mathbb{E}_{X,Y}[xy] - \mathbb{E}_X[x]\mathbb{E}_Y[y] \tag{5}$$

- Multivariate covariance of two multivariate random variables $X \in \mathbb{R}^D, Y \in \mathbb{R}^E$ is:

$$\text{Cov}_{X,Y}[\mathbf{x}, \mathbf{y}] := \mathbb{E}_{X,Y}[\mathbf{x}\mathbf{y}^\top] - \mathbb{E}_X[\mathbf{x}]\mathbb{E}_Y[\mathbf{y}]^\top = \text{Cov}_{Y,X}[\mathbf{y}, \mathbf{x}]^\top \tag{6}$$

- Variance of $\mathbf{x} \in \mathbb{R}^D$ is: $\mathbb{V}_X[\mathbf{x}] = \text{Cov}_X[\mathbf{x}, \mathbf{x}]$.
- Question: what if $\mathbf{x}, \mathbf{y}$ are independent?

- Consider two random variables $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$, then

$$
\begin{aligned}
\mathbb{E}[\mathbf{x} + \mathbf{y}] &= \mathbb{E}[\mathbf{x}] + \mathbb{E}[\mathbf{y}] \\
\mathbb{E}[\mathbf{x} - \mathbf{y}] &= \mathbb{E}[\mathbf{x}] - \mathbb{E}[\mathbf{y}] \\
\mathbb{V}[\mathbf{x} + \mathbf{y}] &= \mathbb{V}[\mathbf{x}] + \mathbb{V}[\mathbf{y}] + 2\mathrm{Cov}[\mathbf{x}, \mathbf{y}] \\
\mathbb{V}[\mathbf{x} - \mathbf{y}] &= \mathbb{V}[\mathbf{x}] + \mathbb{V}[\mathbf{y}] - 2\mathrm{Cov}[\mathbf{x}, \mathbf{y}] \\
\mathbb{E}[\mathbf{A}\mathbf{x} + \mathbf{b}] &= \mathbf{A}\mathbb{E}[\mathbf{x}] + \mathbf{b} \\
\mathbb{V}[\mathbf{A}\mathbf{x} + \mathbf{b}] &= \mathbb{V}[\mathbf{A}\mathbf{x}] = \mathbf{A}\mathbb{V}[\mathbf{x}]\mathbf{A}^\top
\end{aligned} \tag{7}
$$

Parameters of distributions

- How to parameterize a distribution? Use parameter $\boldsymbol{\theta}$, denoting the distribution $p(\mathbf{x}; \boldsymbol{\theta})$.
- There are many formats of distributions: Gaussian (normal) distribution, Bernoulli distribution, categorical distribution.
- E.g., Bernoulli distribution for binary data $\mathbf{x} = \{0, 1\}$,

$$p(\mathbf{x}; \boldsymbol{\theta}) = \theta^{\mathbf{x}}(1-\theta)^{1-\mathbf{x}} = \begin{cases} \theta, & x = 1; \\ 1-\theta, & x = 0. \end{cases} \tag{8}$$

- Categorical distribution for categorical data $\mathbf{x} = 1, 2, 3, \ldots, K$,

$$p(\mathbf{x}; \boldsymbol{\theta}) = \prod_{k=1}^{K} \theta_k^{\mathbf{x}=k} = \begin{cases} \theta_1, & x = 1; \\ \theta_2, & x = 2; \\ \ldots & \ldots \\ \theta_K & x = K. \end{cases} \tag{9}$$

Gaussian distribution

- The pdf of univariate Gaussian distribution (i.e. $x \in \mathcal{N}(x; \boldsymbol{\theta} = \{\mu, \sigma^2\})$) is:

$$p(x; \boldsymbol{\theta} = \{\mu, \sigma^2\}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{10}$$

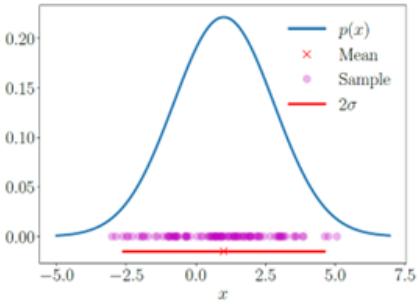
- The pdf of multivariate Gaussian distribution (i.e. $\mathbf{x} \in \mathcal{N}(\mathbf{x}; \boldsymbol{\theta} = \{\boldsymbol{\mu}, \boldsymbol{\Sigma}\})$) is:

$$p(\mathbf{x}; \boldsymbol{\theta} = \{\boldsymbol{\mu}, \boldsymbol{\Sigma}\}) = (2\pi)^{-\frac{D}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right) \tag{11}$$

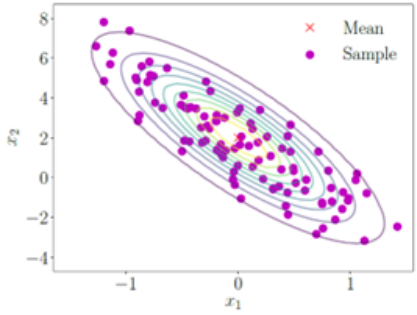
  where $\mathbf{x} \in \mathbb{R}^D$.

- When $\boldsymbol{\mu} = \mathbf{0}, \boldsymbol{\Sigma} = \mathbf{I}$, it is called the *standard normal distribution*.

Visualisation of Gaussian distributions



(a) Univariate (one-dimensional) Gaussian; The red cross shows the mean and the red line shows the extent of the variance.



(b) Multivariate (two-dimensional) Gaussian, viewed from top. The red cross shows the mean and the colored lines show the contour lines of the density.

- For $\mathbf{x} \in \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x), \mathbf{y} \in \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y), a, b \in \mathbb{R}$, we have

$$\mathbf{x} + \mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_x + \boldsymbol{\mu}_y, \boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_y)$$
$$a\mathbf{x} + b\mathbf{y} \sim \mathcal{N}(a\boldsymbol{\mu}_x + b\boldsymbol{\mu}_y, a^2\boldsymbol{\Sigma}_x + b^2\boldsymbol{\Sigma}_y)$$

- For $\mathbf{x} \in \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x), \mathbf{x} \in \mathbb{R}^{D\times 1}, \mathbf{A} \in \mathbb{R}^{M\times D}$, we have

$$\mathbf{A}\mathbf{x} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_x, \mathbf{A}\boldsymbol{\Sigma}_x\mathbf{A}^\top) \tag{12}$$

Sampling from Multivariate Gaussian distributions

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- We assume we can generate samples from a multivariate standard normal as $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
- Use the linear transformation property, if $\mathbf{y} = \mathbf{L}\mathbf{x} + \boldsymbol{\mu}$, where $\mathbf{L}\mathbf{L}^\top = \boldsymbol{\Sigma}$, then $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- The red part refers to the Cholesky decomposition $\mathbf{L}\mathbf{L}^\top = \boldsymbol{\Sigma}$, where $\mathbf{L}$ is the lower-triangular matrix.

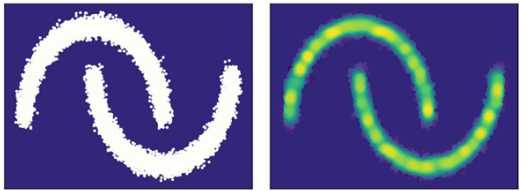
- Given observations $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$, how to *estimate parameters* $\boldsymbol{\theta}$?
- This is called *learning* or *training*.
- In the probabilistic setting, the simplest strategy is to maximize the log-likelihood

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \log p(\mathcal{D}; \boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \sum_{n=1}^{N} \log p(\mathbf{x}_n; \boldsymbol{\theta}) \tag{13}$$

The second $=$ is based on the i.i.d. assumption, i.e.,
$p(\mathcal{D}; \boldsymbol{\theta}) = \prod_{n=1}^{N} p(\mathbf{x}_n; \boldsymbol{\theta})$

Generalization

- We have *true distribution* about observed data (A.), and a fitted model (B.).
- The model assigns probabilities to data, however, it also assigns *some probabilities* to *unseen data*.
- Generalization: a model *generalizes* to regions where no or little data was observed.
- Generalization capability is the key factor is assessing model performance. (test data performance.)



A. Observed data. **B**. A model



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Probabilistic Graphical Models

- PGMs: a graph express the conditional dependence structure between random variables
- E.g., a grey node denotes an observable variable, a white node is a latent variable, and arrows denote conditional dependencies

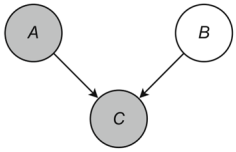


Figure 2: Example graphical model showing the dependencies

- The joint distribution can be written as:

$$p(A, B, C) = P(C|A, B)P(A)P(B) \tag{14}$$

- Mixture model is a simple and very effective model
- E.g., random variable $\mathbf{x}$ - the weight of cats; however, there are female cats and male cats;
- *True* distribution of female cats is Gaussian; *true* distribution of male cates is Gaussian.
- Question: what is the overall distribution over the weight of cats?
- Analysis: $\mathbf{x}$ is observable, the gender of cats is *unknown*, denoted as $\mathbf{z}$. $\mathbf{z} \in \{1 = \text{male}, 0 = \text{female}\}$; the gender $\mathbf{z}$ influences the weight of the cat $\mathbf{x}$

- The joint distribution of weight $\mathbf{x}$ and gender of cat $\mathbf{z}$ can be decomposed as:

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) \tag{15}$$

- $\mathbf{x}$ is real-valued: Gaussian distribution; $\mathbf{z}$ is binary: Bernoulli distribution:

$$\begin{aligned} p(\mathbf{x}, \mathbf{z}) =& \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}(\mathbf{z}), \sigma^2(\mathbf{z}))\text{Bernoulli}(\mathbf{z}|\boldsymbol{\omega}) \\ =& (1-\omega)\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}(0), \sigma^2(0)) + \omega\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}(1), \sigma^2(1)) \end{aligned} \tag{16}$$

where the dependence between $\mathbf{x}$ and $\mathbf{z}$ is written through the mean $\boldsymbol{\mu}(\mathbf{z})$ and the variance $\sigma^2(\mathbf{z})$.

Visualization of Gaussian Mixture Models

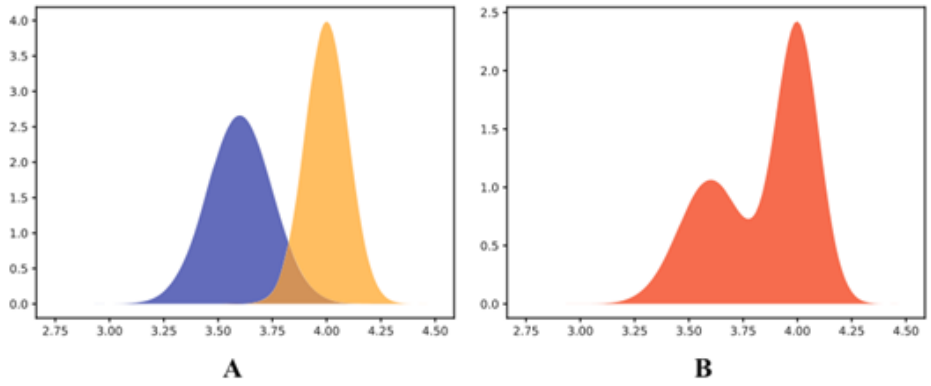


Fig. 2.4 An example of a distribution of weights of cats: (**a**) Gaussians (separate components) for weights of female cats (blue, on the left) and male cats (orange, on the right). (**b**) The marginal distribution of weights of cats, $p(x)$.

General form of Gaussian Mixture Models

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- We may have k different Gaussian distributions
- The general form is:

$$\begin{aligned} p(\mathbf{x};\boldsymbol{\theta}) &= \sum_{\mathbf{z}} \mathcal{N}(\mathbf{x};\boldsymbol{\mu}(\mathbf{z}),\sigma^2(\mathbf{z}))\text{Categorical}(\mathbf{z};\boldsymbol{\omega}) \\ &= \sum_{k=1}^{K} \omega_k \mathcal{N}(\mathbf{x};\boldsymbol{\mu}(k),\sigma^2(k)) \end{aligned} \tag{17}$$

- We can write the log-likelihood term as:

$$\begin{aligned} \log p(\mathbf{x};\boldsymbol{\theta}) &= \log \sum_{k=1}^{K} \omega_k \mathcal{N}(\mathbf{x};\boldsymbol{\mu}(k),\sigma^2(k)) \\ &= \log \sum_{k=1}^{K} \exp\left(\log \omega_k + \log \mathcal{N}(\mathbf{x};\boldsymbol{\mu}(k),\sigma^2(k))\right) \end{aligned} \tag{18}$$

Training Gaussian Mixture Models

- Parameters to be trained: $\{\omega_k, \mu_k, \sigma_k^2\}_{k=1}^K$
- Since we know the log-likelihood function, we can pick optimization method to train the parameters.
  - **Expectation-Maximization**: (i), first calculate the expected probabilities for points belong to specific Gaussian distribution $p(\mathbf{z})$; (ii), update the parameters of Gaussian distributions $\{\omega_k, \mu_k, \sigma_k^2\}_{k=1}^K$.
  - **Gradient-based** optimization methods.





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Continuous Optimization

Visualization understanding of the gradient

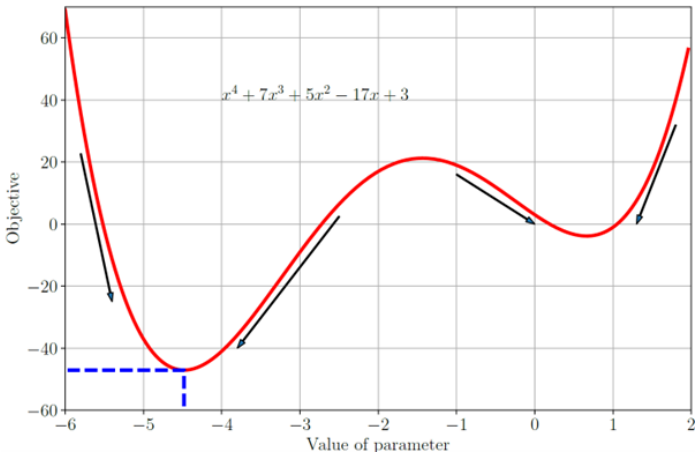


Figure 3: Example objective function. Negative gradients are indicated by arrows, and the global minimum is indicated by the dashed blue line.

Optimization using gradient descent

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- Solving for the minimum of a real-valued function

$$\min_{\mathbf{x}} f(\mathbf{x}) \tag{19}$$

where $f(\cdot)$ is differentiable.

- Procedures of gradient descent:

$$\begin{aligned} \mathbf{x}_1 &= \mathbf{x}_0 - \gamma_0 \left(\frac{\partial}{\partial \mathbf{x}} f(\mathbf{x}) \right) \Big|_{\mathbf{x}=\mathbf{x}_0} \\ &\vdots \\ \mathbf{x}_{i+1} &= \mathbf{x}_i - \gamma_i \left(\frac{\partial}{\partial \mathbf{x}} f(\mathbf{x}) \right) \Big|_{\mathbf{x}=\mathbf{x}_i} \end{aligned} \tag{20}$$

- For suitable step-size γ_i, the sequence $f(\mathbf{x}_0) \geq f(\mathbf{x}_1) \geq f(\mathbf{x}_2) \geq \ldots$ converges to a local minimum.

Example: Gradient descent

- Consider a quadratic function

$$f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^\top \begin{bmatrix} 2 & 1 \\ 1 & 20 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix}^\top \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{21}$$

with gradient $\nabla f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^\top \begin{bmatrix} 2 & 1 \\ 1 & 20 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix}^\top$

- Starting at the initial location $\mathbf{x}_0 = [-3, -1]^\top$, iteratively apply the updating equations to obtain a sequence of estimates that converge to the minimum value.

Visualization of the example

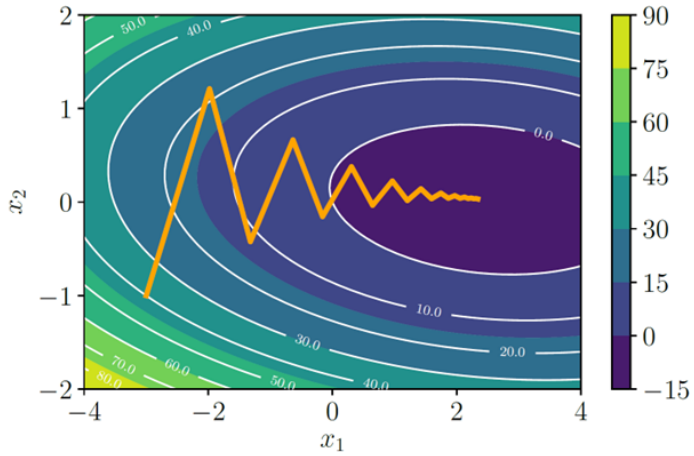


Figure 4: Gradient descent on a two-dimensional quadratic surface (shown as a heatmap).

Conclusion

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In this class, we have learned:

- random variables, common distributions
- properties of random variables and distributions
- Bayes' Theorem
- mean, variance of random variables
- Gaussian distributions
- probabilistic graphical models, Gaussian mixture models
- Continuous optimization

Thank you!